Slides Activity 3:

Study planning

Table of contents

Using simulations to plan a study

If we can posit a plausible data generating mechanism (DGM) for our study, we can use it to ask design questions.

When designing a prospective study, we typically ask "given the effect I hypothesize, how large a sample do I need?"

But when we're using secondary data (e.g., administrative claims, EHR, registry data) the sample size is set by the data.

In this activity we revisit the prenatal vitamins / preterm birth / smoking example from Activities 1–2.

Before you begin

Install the required packages.

```
library(dplyr)
library(tidyr)
library(purrr)
library(ggplot2)
library(kableExtra)
library(scales)
library(sandwich)
```

Step 1. Outline the specific aims.

Context: We are planning a study of the effect of prenatal vitamin supplementation on the risk of preterm birth, worried about confounding by maternal smoking (unmeasured in our planned data source). Before we commit to a data source and sample size, we want to understand how sample size shapes what our study can tell us.

Simulation Objectives:

1. Assuming our hypothesized risk difference is correct, quantify how precision (ASE from both model-based and sandwich standard errors, and ESE) and the proportion of studies whose confidence interval would show a clear difference (power) change across a range of candidate sample sizes.
2. Turning that question around: at a fixed, realistic sample size (as we might encounter with an existing registry or claims database), quantify the smallest effect size we could realistically detect.

Step 2. Determine the appropriate data generating mechanism.

Helper functions

First, create the "helper" functions we will use throughout our simulation.

```
# The expit function is the inverse of the log-odds. It takes in a log-odds (x),
# applies the inverse-logit function and returns a probability
expit <- function(x) {
  return(exp(x) / (1 + exp(x)))
}

# The balancing intercept function, so we can re-derive the intercept whenever we
# change other parameters
derive_intercept <- function(py, prob_prenatal, prob_smoke,
                             or_smoke, or_prenatal){

  # marginal probability of treatment
  prob_prenatal <- prob_prenatal[1] * (1-prob_smoke) + prob_prenatal[2] * prob_smoke

  -log(1/py - 1) - or_prenatal*prob_prenatal - or_smoke*prob_smoke
}
```

Function for DGM

This is the same DGM from Activities 1–2.

```
dgm <- function(simid = 0,
                n = 500,
                prob_smoke = 0.2,
                prob_prenatal = c(0.85, 0.65),
                or_prenatal = log(0.8), # increasing the effect
                or_smoke = log(3),
                prob_preterm = 0.1){

  # Dynamically derive the intercept term
  y_intercept <- derive_intercept(py = prob_preterm,
                                  prob_prenatal = prob_prenatal,
                                  prob_smoke = prob_smoke,
                                  or_smoke = or_smoke,
                                  or_prenatal = or_prenatal)
```

```
  dplyr::tibble(
    # Simulation number
    simid = simid,

    # Create n participants
    id = 1:n,

    # Randomly select smoking
    smoke = rbinom(n = n, size = 1, prob = prob_smoke),

    # Generate potential outcome under no treatment
    preterm_x0 = rbinom(n = n, size = 1,
                        prob = expit(y_intercept + (or_smoke * smoke))),

    # Generate potential outcome under treatment
    preterm_x1 = rbinom(n = n, size = 1,
                        prob = expit(y_intercept + (or_smoke * smoke) + or_prenatal)),

    # Generate observed treatment conditional on smoking
    p_prenatal = prob_prenatal[smoke + 1],
    p_prenatal_margin = prob_prenatal[1]*(1-prob_smoke) + prob_prenatal[2]*(prob_smoke),
    prenatal = rbinom(n = n, size = 1, prob = p_prenatal),

    # Assign actual outcome conditional on observed treatment
    preterm = preterm_x0*(1-prenatal) + preterm_x1*(prenatal),

    # Known IPT weight
    ipw = dplyr::if_else(
      prenatal == 1,
      p_prenatal_margin / p_prenatal,
      p_prenatal_margin / (1 - p_prenatal)
    )
  )
}
```

Set the seed

ALWAYS before any random number generation!

```
set.seed(918273)
```

The hypothesized effect

Study planning mainly *before* we have data, so instead of an empirical "truth" recovered from a huge dataset, we plug in our best guess (from prior literature, pilot data, or clinical judgment) for each DGM parameter. We can still generate one very large sample under those hypothesized parameters just to confirm what marginal risk difference they imply.

```
truth <- dgm(n = 10000000)
true_rd <- mean(truth$preterm_x1) - mean(truth$preterm_x0)
true_rd
```

```
[1] -0.0218565
```

Step 3. Define the estimand of the analysis.

As in Activities 1–2, our estimand is the marginal risk difference, $E[Y^{x=1}] - E[Y^{x=0}]$: the risk of preterm birth had *everyone* in the target population taken prenatal vitamins vs. had *no one* taken them.

Step 4. Identify the methods to be implemented in the study.

We will estimate the IPT-weighted risk difference (known weights), like Activity 2. Alongside the model-based standard error, we'll compute the robust sandwich standard error, which accounts for using (known) weights.

Step 5. Specify the performance measures.

Precision. As in Activity 2, we'll track a few measures of how precisely we can pin down the risk difference:

- **ASE, model-based**: the average, across simulated samples, of the model-based standard error reported by the regression — what our software *tells us* about precision in a typical single sample.
- **ASE, sandwich**: the same idea, but averaging the robust sandwich standard error instead.
- **ESE** (empirical standard error): the standard deviation of the estimated risk differences across simulated samples. Captures the sampling variability of the estimator for the risk difference.

Power. Here we define power simply in terms of the confidence interval: the proportion of simulated samples whose 95% confidence interval (built from the model-based standard error) for the risk difference excludes zero (the value that would mean "no difference").

$$Power \approx P\Big(0 \notin \big[\widehat{RD} - 1.96 \times \widehat{SE}(\widehat{RD}),\ \widehat{RD} + 1.96 \times \widehat{SE}(\widehat{RD})\big]\Big)$$

What proportion of studies of each size N would the confidence interval exclude 0, if our hypothesized effect is correct?

Simulating across a range of sample sizes

Function to simulate one sample-size scenario

This function draws `nsim` samples at a given sample size `n`, fits the IPTW model to each, and returns the estimated risk difference, the model-based and sandwich standard errors, whether the 95% CI excludes zero, and whether the 95% CI includes the true risk difference.

```
run_scenario <- function(n, nsim = 300) {
  purrr::map_dfr(1:nsim, function(i) {
    dat <- dgm(simid = i, n = n)

    unadj <- glm(preterm ~ prenatal, data = dat, family = binomial(link = "identity"))
    adj_known <- glm(preterm ~ prenatal, data = dat,
                     weights = ipw, family = quasibinomial(link = "identity"))

    dplyr::tibble(
      estimator = c("Unadj.", "Adj. (Model-based SE)", "Adj. (Robust SE)"),

      # extract risk different from the models
      rd = c(coef(unadj)[2], coef(adj_known)[2], coef(adj_known)[2]),

      # extract the standard errors from the models
      se = c(sqrt(diag(vcov(unadj)))[2],
             sqrt(diag(vcov(adj_known)))[2],
             sqrt(diag(sandwich::vcovHC(adj_known)))[2])
    )
    }) |>
    dplyr::mutate(
      lcl = rd - 1.96*se,
      ucl = rd + 1.96*se,
      # Does this replicate's 95% CI exclude zero?
```

```
      excludes_zero = (lcl > 0) | (ucl < 0),
      # Does this replicate's CI include the truth?
      includes_truth = lcl <= true_rd & true_rd <= ucl
    )
}
```

Run this across a grid of candidate sample sizes

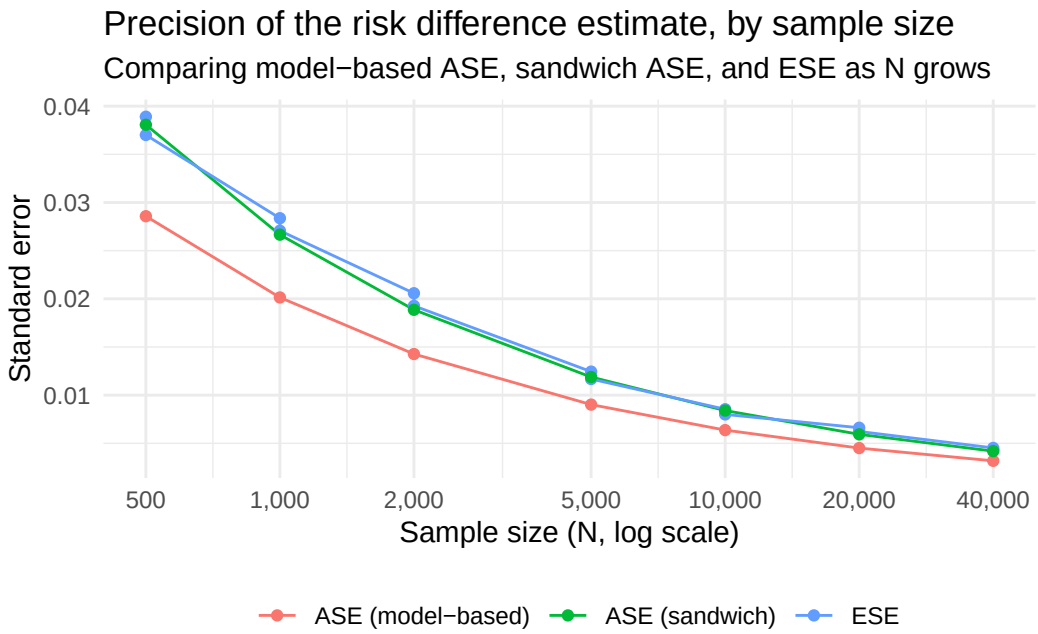
```
n_grid <- c(500, 1000, 2000, 5000, 10000, 20000, 40000)

results_by_n <- purrr::map_dfr(n_grid, function(n) {
  reps <- run_scenario(n, nsim = 300)
  reps |>
    dplyr::summarize(
      n = n,
      power = mean(excludes_zero),
      ase = sqrt(mean(se^2)),
      ese = sd(rd),
      coverage = mean(includes_truth),
      .by = estimator
  )
})

results_by_n |>
  dplyr::arrange(estimator, n) |>
  kableExtra::kable(col.names = c("Estimator", "N", "Power",  "ASE", "ESE", "95% CI coverage
                    digits = c(0, 3, 4, 4, 4, 3))
```

Estimator	N	Power	ASE	ESE	95% CI coverage
Adj. (Model-based SE)	500	0.2167	0.0286	0.0370	0.850
Adj. (Model-based SE)	1000	0.2500	0.0201	0.0271	0.850
Adj. (Model-based SE)	2000	0.3600	0.0143	0.0193	0.877
Adj. (Model-based SE)	5000	0.6200	0.0090	0.0117	0.873
Adj. (Model-based SE)	10000	0.9067	0.0064	0.0080	0.877
Adj. (Model-based SE)	20000	0.9867	0.0045	0.0062	0.840
Adj. (Model-based SE)	40000	1.0000	0.0032	0.0043	0.843
Adj. (Robust SE)	500	0.0733	0.0381	0.0370	0.960
Adj. (Robust SE)	1000	0.1167	0.0267	0.0271	0.933
Adj. (Robust SE)	2000	0.1667	0.0189	0.0193	0.940

Estimator	N	Power	ASE	ESE	95% CI coverage
Adj. (Robust SE)	5000	0.4433	0.0119	0.0117	0.933
Adj. (Robust SE)	10000	0.7467	0.0084	0.0080	0.957
Adj. (Robust SE)	20000	0.9700	0.0059	0.0062	0.957
Adj. (Robust SE)	40000	1.0000	0.0042	0.0043	0.933
Unadj.	500	0.2367	0.0398	0.0389	0.900
Unadj.	1000	0.4633	0.0280	0.0284	0.830
Unadj.	2000	0.7500	0.0198	0.0206	0.717
Unadj.	5000	0.9867	0.0125	0.0124	0.330
Unadj.	10000	1.0000	0.0089	0.0085	0.057
Unadj.	20000	1.0000	0.0063	0.0066	0.003
Unadj.	40000	1.0000	0.0044	0.0045	0.000

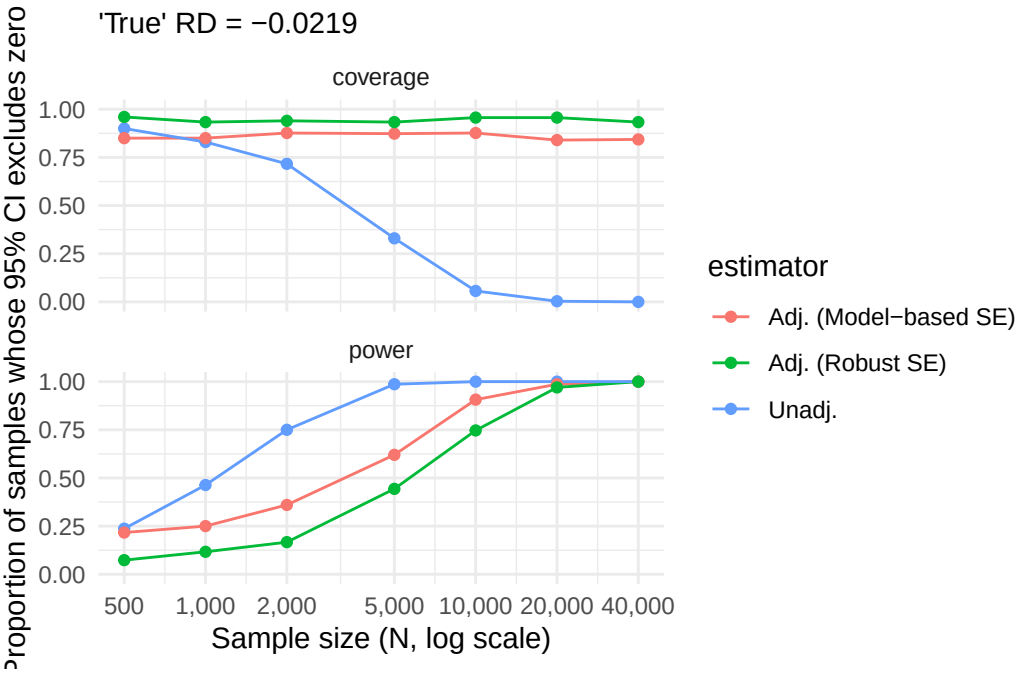
How does N affect precision (ASE, sandwich ASE, and ESE)?

```
results_by_n |>
  tidyr::pivot_longer(cols = c(ase, ese), names_to = "measure", values_to = "value") |>
  dplyr::mutate(
    measure = dplyr::case_when(
      measure == "ese" ~ "ESE",
      measure == "ase" & estimator == "Adj. (Model-based SE)" ~ "ASE (model-based)",
      measure == "ase" & estimator == "Adj. (Robust SE)" ~ "ASE (sandwich)")) |>
  distinct(n, value, measure) |>
  dplyr::filter(!is.na(measure)) |>
  ggplot(aes(x = n, y = value, color = measure)) +
  geom_line() +
  geom_point() +
  scale_x_log10(breaks = n_grid, labels = scales::label_comma()) +
  theme_minimal() +
  labs(
    x = "Sample size (N, log scale)",
    y = "Standard error",
    color = "",
    title = "Precision of the risk difference estimate, by sample size",
    subtitle = "Comparing model-based ASE, sandwich ASE, and ESE as N grows"
  ) +
  theme(legend.position = "bottom")
```



How does N affect power?

```
results_by_n |>
  pivot_longer(cols = c(power, coverage)) |>
  ggplot(aes(x = n, y = value, color = estimator, group = estimator)) +
  geom_line() +
  geom_point() +
  facet_wrap(~name, nrow = 2) +
  scale_x_log10(breaks = n_grid, labels = scales::label_comma()) +
  scale_y_continuous(limits = c(0, 1)) +
  theme_minimal() +
  labs(
    x = "Sample size (N, log scale)",
    y = "Proportion of samples whose 95% CI excludes zero",
    subtitle = paste0("'True' RD = ", round(true_rd, 4))
  )
```



The unadjusted and adjusted with model-based SEs technically have better power, but only because the unadjusted is biased further from the null and the model-based SEs are small! Coverage still gives a better picture, **especially when our goal is estimation, not testing.**

What effect could I detect with the data I have?

A planning question could be: "given the N in this data source (and my design choices), what's the smallest effect I could realistically detect?"

Or, "given my N, and my hypothesized effect, what is the precision of my estimate?"

We can use the same data generating mechanism, just looping over different parameters. The first, again, is a question of power. The second is just an approach to understand the type of precision we can have

Function to simulate one effect-size scenario

```
run_effect_scenario <- function(n, or_prenatal, nsim = 250) {
  purrr::map_dfr(1:nsim, function(i) {
    dat <- dgm(simid = i, n = n, or_prenatal = or_prenatal)
```

```
    fit <- glm(preterm ~ prenatal, data = dat,
               weights = ipw, family = quasibinomial(link = "identity"))

    # below we are only using the sandwich SEs because now we know that's the more
    # correct way to go!
    dplyr::tibble(
      rd = coef(fit)[2],
      se = sqrt(diag(sandwich::vcovHC(fit)))[2]
    )
  }) |>
    dplyr::mutate(
      ci_width = (rd + 1.96*se) - (rd - 1.96*se),
      excludes_zero = (rd - 1.96*se > 0) | (rd + 1.96*se < 0)
    )
}
```

We use the sandwich standard error directly here, since we know it more accurately estimates the empirical standard error.

Translating a hypothesized risk difference into the DGM

Our DGM generates the treatment effect on the log-odds scale (`or_prenatal`), but the effect we actually want to hypothesize about is the marginal risk difference. Below root solves for the OR necessary to target a specified risk difference.

```
implied_rd <- function(or_prenatal, or_smoke = log(3), prob_smoke = 0.2,
                       prob_prenatal = c(0.85, 0.65), prob_preterm = 0.1) {
  y_intercept <- derive_intercept(py = prob_preterm, prob_prenatal = prob_prenatal,
                                  prob_smoke = prob_smoke, or_smoke = or_smoke,
                                  or_prenatal = or_prenatal)
  p_x0 <- (1 - prob_smoke) * expit(y_intercept) + prob_smoke * expit(y_intercept + or_smoke)
  p_x1 <- (1 - prob_smoke) * expit(y_intercept + or_prenatal) + prob_smoke * expit(y_intercep

  p_x1 - p_x0
}

or_prenatal_for_rd <- function(target_rd) {
  # the below takes a target risk difference and finds the log-odds ratio that would produce
  # root solves such that the output of implied_rd minus the target RD equals 0
  uniroot(function(lor) implied_rd(lor) - target_rd, interval = log(c(0.001, 10)))$root
}
```

Run this across a grid of hypothesized effect sizes, at a fixed sample size

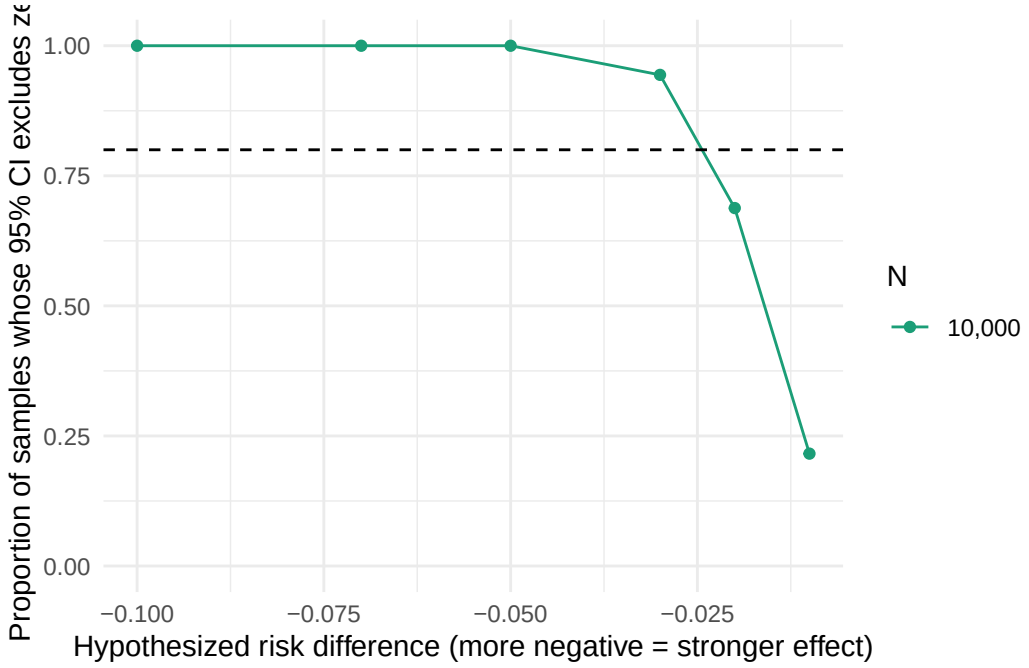
```
rd_grid <- c(-0.01, -0.02, -0.03, -0.05, -0.07, -0.10)
fixed_n <- 10000

detectable_effect <- purrr::map_dfr(
  fixed_n,
  function(n) {
    purrr::map_dfr(rd_grid,
                   function(target_rd) {
                     reps <- run_effect_scenario(n,
                                                 # below calculates the OR for a given RD
                                                 or_prenatal = or_prenatal_for_rd(target_rd),
                                                 nsim = 250)
                     dplyr::tibble(n = n,
                                   rd = target_rd,
                                   power = mean(reps$excludes_zero))
                     }
                   )
    }
  )

detectable_effect |>
  kableExtra::kable(col.names = c("N", "Hypothesized risk difference", "Power"), digits = 3)
```

N	Hypothesized risk difference	Power
10000	-0.01	0.216
10000	-0.02	0.688
10000	-0.03	0.944
10000	-0.05	1.000
10000	-0.07	1.000
10000	-0.10	1.000

Plot: power vs. hypothesized effect size, at N = 10,000

```
ggplot(detectable_effect, aes(x = rd, y = power, color = factor(n))) +
  geom_line() +
  geom_point() +
```

```
  geom_hline(yintercept = 0.8, linetype = "dashed") +
  scale_y_continuous(limits = c(0, 1)) +
  scale_color_manual(values = c("#1b9e77", "#7570b3"), labels = scales::comma(fixed_n)) +
  theme_minimal() +
  labs(
    x = "Hypothesized risk difference (more negative = stronger effect)",
    y = "Proportion of samples whose 95% CI excludes zero",
    color = "N"
  )
```



The minimum detectable effect size at 80% power

```
mdes <- detectable_effect |>
  dplyr::arrange(n, power) |>
  dplyr::summarize(
    min_detectable_rd = approx(x = power, y = rd, xout = 0.8)$y,
    .by = n
  )

mdes |>
  kableExtra::kable(col.names = c("N", "Smallest detectable risk difference (80% power)"), di
```

N	Smallest detectable risk difference (80% power)
10000	-0.024